CONJECTURING · CONSECUTIVE SUMS · DIVISORS AND CONGRUENCES

Chains of Consecutive Numbers

A ladder of twenty-four conjectures, six of which fall, each preceded by the examples that suggest it

Some numbers are sums of consecutive positive integers, as in $9 = 2 + 3 + 4$, $11 = 5 + 6$ and $18 = 3 + 4 + 5 + 6$, and asking which numbers have the property, how long the sums can be and in how many ways a single number can be written is a seed problem for learning to conjecture, taken here from Mason, Burton and Stacey [1]. Their first guesses are reached by specialising and testing. We develop prime factoring the counting of divisors; reading congruence classes; combining congruences with the Chinese remainder theorem and the least common multiple; separating the power of 2 from the primitive part; pairing objects off by a symmetry; and counting lattice points under a curve.

0 · DEFINITIONS

The vocabulary, with examples

Chain. A chain of N is a run of $k \geq 2$ consecutive positive integers whose sum is N. The number 15 has exactly three chains, $15 = 7 + 8 = 4 + 5 + 6 = 1 + 2 + 3 + 4 + 5$, and the one-term sum $15 = 15$ is not counted, nor is the run $-6 + \cdots + 8$, which reaches 15 through zero and negative terms. The number 16 has no chain at all.

Start, end, length and middle term. The chain $4 + 5 + 6$ has start 4, end 6 and length 3. A chain of odd length has a middle term, here $5 = 15/3$, and a chain of even length has none, as with $7 + 8$, whose mean is 7.5. The abbreviation a–b stands for $a + (a + 1) + \cdots + b$, so 4–6 is the chain just described.

Degeneracy. The degeneracy $r(N)$ is the number of distinct chains of N, so $r(15) = 3$, $r(9) = 2$ (from $4 + 5$ and $2 + 3 + 4$), $r(3) = 1$ and $r(16) = 0$. Two exclusions are fixed from the start. The one-term sum is not a chain, since counting it would add 1 to every value, and runs through zero or negatives such as $9 = -1 + 0 + 1 + 2 + 3 + 4$ are not chains either, though Conjecture 22 counts them separately. Because a given length can occur at most once in a number, the degeneracy is also the number of different chain lengths that N allows.

Triangular numbers. $T_k = 1 + 2 + \cdots + k$, so $T_3 = 6$, $T_4 = 10$, $T_5 = 15$. The chain 1–k is the chain of length k with the smallest possible sum, so T_k is the smallest number that has a chain of length k.

Odd divisors. $\tau_{\text{odd}}(N)$ counts the odd divisors of N (including 1 and, when N is odd, N itself). The number 12 has odd divisors $1, 3$, so $\tau_{\text{odd}}(12) = 2$, and 45 has $1, 3, 5, 9, 15, 45$, so $\tau_{\text{odd}}(45) = 6$.

Primitive numbers and scaled instances. A number is *primitive* when it is odd. Every N is uniquely $N = 2^j m$ with m primitive, and then m is the *primitive part* of N and j its *doubling depth*, so that N is the j-fold doubling of m. For instance 15 is primitive ($j = 0$), $12 = 2^2 \cdot 3$ is the second doubling of the primitive number 3, $40 = 2^3 \cdot 5$ is the third doubling of 5, and a power of 2 has primitive part 1. A scaled instance keeps the degeneracy of its primitive part while its chains change shape: $r(3) = r(6) = r(12) = 1$, with chains $1{+}2$, then $1{+}2{+}3$, then $3{+}4{+}5$.

N	chains a–b	r	N	chains a–b	r
1	*none*	0	17	8–9	1
2	*none*	0	18	5–7, 3–6	2
3	1–2	1	19	9–10	1
4	*none*	0	20	2–6	1
5	2–3	1	21	10–11, 6–8, 1–6	3
6	1–3	1	22	4–7	1
7	3–4	1	23	11–12	1
8	*none*	0	24	7–9	1
9	4–5, 2–4	2	25	12–13, 3–7	2
10	1–4	1	26	5–8	1
11	5–6	1	27	13–14, 8–10, 2–7	3
12	3–5	1	28	1–7	1
13	6–7	1	29	14–15	1
14	2–5	1	30	9–11, 6–9, 4–8	3
15	7–8, 4–6, 1–5	3	31	15–16	1
16	*none*	0	32	*none*	0

The entry a–b stands for $a+(a+1)+\cdots+b$, and *none* marks a number with no chain.

I · WHICH NUMBERS

Noticing the first families

NOTICE

$3=1+2,\quad 5=2+3,\quad 7=3+4,\quad 9=4+5,\quad 11=5+6,\quad 13=6+7.$

Pause. In each line the two terms differ by one and sit on either side of $N/2$, so the smaller term should be the whole-number part of $N/2$.

Conjecture 1 — Specialise · Stands

Every odd number $N\geq 3$ is the sum of two consecutive numbers.

TEST. The identity $N=a+(a+1)=2a+1$ gives $a=(N-1)/2$, which is a positive integer for every odd $N\geq 3$, and the table shows it all the way to $31=15+16$.

See all conjecture tests in python.

NOTICE

Among $1,2,3,4,5$ the numbers that have chains are 3 and 5, and the numbers with none are $1,2,4$. The numbers with chains are the odd ones.

Pause. Five cases is a small sample, and the useful question is whether the even numbers fail because something prevents a chain or because nothing larger has been tried.

Conjecture 2 — Specialise · Falls

A number has a chain only if it is odd.

TEST. The table already contains $6=1+2+3$, $10=1+2+3+4$, $12=3+4+5$, $14=2+3+4+5$, and also $18,20,22,24$, so the conjecture fails at the first even number past 4. Its failure sends attention to the chains with more than two terms, which the odd-number argument of Conjecture 1 never touches.

NOTICE

Multiples of 3: $6=1+2+3,\ 9=2+3+4,\ 12=3+4+5,\ 15=4+5+6,\ 18=5+6+7,\ 21=6+7+8.$
Multiples of 5: $15=1+2+3+4+5,\ 20=2+\cdots+6,\ 25=3+\cdots+7,\ 30=4+\cdots+8.$
The middle term of each run is $N/3$ or $N/5$, and 3 and 5 themselves refuse: $3=0+1+2$ and $5=-1+0+1+2+3$ need a zero or a negative term, as does $10=0+1+2+3+4$.

Pause. Write a multiple of 3 as $3F$ and ask what the three terms are, and then ask for which F the smallest of them is positive.

Conjecture 3 — Factor · Stands, repairing 2

If $N=(2K+1)F$ then $N=(F-K)+\cdots+F+\cdots+(F+K)$, a run of $2K+1$ terms centred on F, and every term is positive exactly when $F\geq K+1$, that is, when $N\geq(2K+1)(K+1)=T_{2K+1}$.

TEST. Each of the $2K+1$ terms deviates from F by an amount $-K,\ldots,K$ and the deviations cancel in pairs, so the sum is $(2K+1)F$; the smallest term $F-K$ is positive when $F\geq K+1$. Taking $15=3\cdot 5$ gives two chains, $4+5+6$ from $K=1,F=5$ and $1+\cdots+5$ from $K=2,F=3$, while the odd factor 15 itself would need the run $-6,\ldots,8$, which is no chain.

The threshold T_{2K+1} is the reason a multiple of 3 gets its chain only from 6 and a multiple of 5 only from 15, so every number with an odd factor above 1 has a candidate chain, and the failures at 10 and 14 show that the candidate does not always survive.

NOTICE

$10 = 0+1+2+3+4 \;\to\; 1+2+3+4, \qquad 5 = -1+0+1+2+3 \;\to\; 2+3,$
$14 = -1+0+1+2+3+4+5 \;\to\; 2+3+4+5, \qquad 21 = 0+1+2+3+4+5+6 \;\to\; 1+2+3+4+5+6.$

Pause. Each run that fell short of positivity was repaired by deleting the zero and cancelling every negative term against its positive twin, and the question is how many terms survive, and whether the survivors always begin where the pattern suggests.

Conjecture 4 — **Pair off · Stands**

If $d = 2K+1 > 1$ divides N and $F = N/d \le K$, the non-positive terms cancel against positive ones and leave the chain $(K-F+1)+\cdots+(K+F)$ of length $2F = 2N/d$. Every odd divisor $d > 1$ of N therefore yields exactly one chain, of length d when $N/d > (d-1)/2$ and of length $2N/d$ otherwise.

TEST. For $14 = 7\cdot 2$ we have $K = 3, F = 2$, the run $-1,\dots,5$ loses $-1, 0, 1$ and keeps $2,\dots,5$, a chain of length $4 = 2F$ starting at $K - F + 1 = 2$; the other examples give lengths $4, 2, 6$ in the same way.

The cancellation is a symmetry of the run about its centre, and it is the first appearance of the pairing habit, which returns in Conjectures 11, 21 and 22. For $N = 15$ the factor $d = 15$ now yields $7+8$, the two-term chain of Conjecture 1, so the odd numbers of that conjecture are the case $d = N$.

NOTICE

For $N = 21$ double it to 42 and list the factor pairs, $1\cdot 42,\ 2\cdot 21,\ 3\cdot 14,\ 6\cdot 7$. The pairs $(2,21), (3,14), (6,7)$ give $a = \frac{m-k+1}{2} = 10,\ 6,\ 1$ and so the chains $10+11,\ 6+7+8,\ 1+\cdots+6$.
For $N = 12$ double it to 24 and list $1\cdot 24,\ 2\cdot 12,\ 3\cdot 8,\ 4\cdot 6$, and only $(3,8)$ gives a chain, $3+4+5$.

Pause. In 21 every factor pair mixes an odd and an even number, and in 24 the pairs $(2,12)$ and $(4,6)$ are the ones that produce nothing, which suggests asking what the sum of k terms from a is when written out.

Conjecture 5 — **Double and factor · Stands**

A chain of k terms from a sums to N exactly when $2N = k(2a+k-1)$. Chains correspond one to one with factorisations $2N = k\cdot m$ with $2 \le k < m$ and k, m of opposite parity, with $a = (m-k+1)/2$.

TEST. The sum of k terms from a is $ka + k(k-1)/2$, and doubling clears the fraction, giving $k(2a+k-1)$. The two factors k and $2a+k-1$ differ by $2a-1$, which is odd, so one factor is even and the other odd, and $m > k$ because $a \ge 1$.

The odd factor divides N, and which factor is the smaller one decides the parity of the length, which is the content of Conjectures 3 and 4.

II · HOW MANY CHAINS

Counting by factoring

NOTICE

15: odd divisors $1,3,5,15$, chains $7{+}8,\ 4{+}5{+}6,\ 1{+}\ldots{+}5$. (3 chains)
21: odd divisors $1,3,7,21$, chains $10{+}11,\ 6{+}7{+}8,\ 1{+}\ldots{+}6$. (3 chains)
45: odd divisors $1,3,5,9,15,45$, chains starting $22,\ 14,\ 7,\ 5,\ 1$. (5 chains)
9: odd divisors $1,3,9$. (2 chains) 12: odd divisors $1,3$. (1 chain)

Pause. Compare the number of chains with the number of odd divisors in every line, and then ask what r is for $N = 3^2\cdot 5 = 45$ written as a product of prime powers.

Conjecture 6 — **Factor · Stands**

The degeneracy is $r(N) = \tau_{\text{odd}}(N) - 1$. For $N = 2^j \prod p_i^{e_i}$ with the p_i odd primes, $r(N) = \prod(e_i+1) - 1$.

TEST. By Conjecture 5 chains correspond to the odd divisors d of N, and the divisor $d = 1$ gives only the one-term

chain N itself. Divisors of a prime power p^e number $e+1$ and the count of divisors is multiplicative over coprime parts, so $45 = 3^2 \cdot 5$ has $3 \cdot 2 - 1 = 5$ chains as the table shows.

NOTICE

No chains: $1,\ 2,\ 4,\ 8,\ 16,\ 32.$ Chains: $3,\ 6,\ 12,\ 24,\ 48,\ \ldots$ (every $3 \cdot 2^j$).

Pause. The empty rows are the powers of 2, and the multiples of 3 by powers of 2 are never empty, so the question is what a power of 2 lacks that $3 \cdot 2^j$ has.

Conjecture 7 — Factor · Stands

N has no chain if and only if N is a power of 2.

TEST. A power of 2 has the single odd divisor 1, so $r = \tau_{\text{odd}} - 1 = 0$ by Conjecture 6, and any N with an odd divisor above 1 has $r \geq 1$.

NOTICE

Primes with exactly one chain: $3 = 1+2,\ 5 = 2+3,\ 7 = 3+4,\ 11 = 5+6,\ 13 = 6+7,\ 17 = 8+9,\ 19 = 9+10.$

Pause. Every prime in the list has the single chain of two terms, and the next question is whether any composite number can have exactly one chain.

Conjecture 8 — Specialise · Falls

N has exactly one chain if and only if N is an odd prime.

TEST. The numbers $6 = 1+2+3$, $10 = 1+2+3+4$, $12 = 3+4+5$, $14 = 2+3+4+5$, $20 = 2+\cdots+6$, 22 and 24 each carry exactly one chain and none is an odd prime. Written as $2 \cdot 3,\ 2 \cdot 5,\ 4 \cdot 3,\ 2 \cdot 7,\ 4 \cdot 5,\ 2 \cdot 11,\ 8 \cdot 3$ they share a shape, and the shape is the repair.

Conjecture 9 — Factor · Stands, repairing 8

N has exactly one chain if and only if $N = 2^j p$ with p an odd prime.

TEST. By Conjecture 6, $r(N) = 1$ means $\tau_{\text{odd}}(N) = 2$, so the primitive part m has exactly two divisors, which is the definition of a prime.

NOTICE

Doubling a number with three chains: $15,\ 30,\ 60,\ 120$ all have $r = 3$, with chains $15:\ 7{+}8,\ 4{+}5{+}6,\ 1{+}\ldots{+}5$
$30:\ 9{+}10{+}11,\ 6{+}\ldots{+}9,\ 4{+}\ldots{+}8$
$60:\ 19{+}20{+}21,\ 10{+}\ldots{+}14,\ 4{+}\ldots{+}11$ $\quad 120:\ 39{+}40{+}41,\ 22{+}\ldots{+}26,\ 1{+}\ldots{+}15.$
The lengths drift, $\{2,3,5\}$, $\{3,4,5\}$, $\{3,5,8\}$, $\{3,5,15\}$, while the count stays put. Likewise the single chain of $3, 6, 12, 24, 48$ is $1{+}2$, $1{+}2{+}3$, $3{+}4{+}5$, $7{+}8{+}9$, $15{+}16{+}17$.

Pause. The chains of 15 and 120 differ in length and in position, and the question is which feature of the number is invisible to the count, now that 15, 30, 60, 120 are the primitive number 15 and its first three doublings.

Conjecture 10 — Factor · Stands

Every scaled instance has the degeneracy of its primitive part, $r(2^j m) = r(m)$, so that doubling preserves degeneracy, $r(2N) = r(N)$. Moreover $r = 2$ if and only if $m = p^2$, and $r = 3$ if and only if $m = pq$ or $m = p^3$ for distinct odd primes p, q. The least N with at least r chains, for $r = 1, \ldots, 12$, are $3, 9, 15, 45, 45, 105, 105, 225, 315, 315, 315, 945$.

TEST. Conjecture 6 makes r a function of m, and $\tau_{\text{odd}} = 3$ forces $m = p^2$ while $\tau_{\text{odd}} = 4$ forces $m = pq$ or p^3, since 4 factors only as $2 \cdot 2$ or 4.

The record-setters are all primitive, an even number gaining no chains from its factor of 2, and each is the smallest primitive number with the required divisor count, such as $45 = 3^2 \cdot 5$ for $\tau_{\text{odd}} = 6$ and $105 = 3 \cdot 5 \cdot 7$ for $\tau_{\text{odd}} = 8$.

NOTICE

$r = 2$: $9,\ 18,\ 25,\ 36.$ $\quad r = 3$: $15,\ 21,\ 27,\ 30.$ $\quad r = 5$: $45,\ 75.$

Primitive parts: $9 = 3^2$, $18 \to 9$, $25 = 5^2$, $36 \to 9$ are squares; $15,\ 21,\ 27 = 3^3,\ 30 \to 15,\ 45,\ 75$ are not.

Pause. In the second line the numbers with an even number of chains have a square primitive part, and the question is why a square should be special for divisors.

Conjecture 11 — Pair off · Stands

$r(N)$ is even if and only if the primitive part m is a perfect square. Hence N has an even number of chains exactly when $N = 2^j m^2$.

TEST. The divisors of m pair off as $d \leftrightarrow m/d$, and a divisor is paired with itself only when $d^2 = m$. A square has one unpaired divisor, so τ_{odd} is odd and $r = \tau_{\text{odd}} - 1$ is even, while a non-square has all divisors paired, so τ_{odd} is even and r is odd. This is the same symmetry as in Conjecture 4, now acting on divisors instead of on terms.

III · HOW LONG

Lengths, congruences and the power of 2

NOTICE

Length 3: $6,\ 9,\ 12,\ 15,\ 18,\ 21, \dots$	step 3, first term 6
Length 4: $10,\ 14,\ 18,\ 22,\ 26,\ 30, \dots$ $(1{+}2{+}3{+}4,\ 2{+}\dots{+}5,\ 3{+}\dots{+}6, \dots)$	step 4, first term 10
Length 5: $15,\ 20,\ 25,\ 30, \dots$	step 5, first term 15
Length 6: $21,\ 27,\ 33,\ 39,\ 45, \dots$ $(1{+}\dots{+}6,\ 2{+}\dots{+}7,\ 3{+}\dots{+}8, \dots)$	step 6, first term 21

Pause. Each row is an arithmetic progression whose step equals its length and whose first term is the sum $1 + \cdots + k$, and sliding the window one place to the right adds k to the sum. The question is what residue class mod k the progression occupies, and whether odd and even k behave alike.

Conjecture 12 — Congruence class · Stands

A chain of length k exists if and only if $N \geq T_k$ and $N \equiv T_k \pmod{k}$, that is, $N = T_k + (a-1)k$ for the start a. For odd k this says $k \mid N$, and for even k it says $N \equiv k/2 \pmod{k}$.

TEST. Moving the window from start a to $a+1$ raises every one of the k terms by 1, so N rises by exactly k, and the first window, $a = 1$, sums to T_k. Since $T_k = k \cdot \frac{k+1}{2}$ the residue is 0 when k is odd, and when k is even $T_k = \frac{k}{2}(k+1) = \frac{k}{2} \cdot k + \frac{k}{2} \equiv \frac{k}{2} \pmod{k}$.

k	2	3	4	5	6	7	8
$N \bmod k$	1	0	2	0	3	0	4
first $N = T_k$	3	6	10	15	21	28	36

A proportion $1/k$ of the integers beyond T_k therefore carries a chain of length k, and the multiples of 3 and 5 of the first rungs are simply the rows $k = 3$ and $k = 5$.

NOTICE

Two lengths in one number: 18 has lengths 3 and 4 $(5{+}6{+}7$ and $3{+}\dots{+}6)$; 30 has $3, 4, 5$; 15 has $2, 3, 5$; 21 has $2, 3, 6$; 75 has $2, 3, 5, 6, 10$.

Two lengths that have not yet met in the table: 2 and 4; 4 and 6.

Pause. Every pair of lengths that was looked for has turned up together in some number, so the natural guess is that any two lengths coexist once N is large enough, and the pair $2, 4$ is worth hunting for before believing it.

Conjecture 13 — Congruence class · Falls

For any two lengths $k, k' \geq 2$ there are infinitely many N having a chain of each length.

TEST. A chain of length 2 needs $N \equiv 1 \pmod 2$, so N is odd, and a chain of length 4 needs $N \equiv 2 \pmod 4$, so N is even. The two conditions contradict each other, and no value of N, however large, satisfies both. The pair $4, 6$ fails in the same way, since $N \equiv 2 \pmod 4$ is even while $N \equiv 3 \pmod 6$ is odd.

NOTICE

Compatible pairs, found by solving the two congruences together:
lengths $3, 4$: $N \equiv 0 \pmod 3$ and $N \equiv 2 \pmod 4$ give $N \equiv 6 \pmod{12}$, so $18,\ 30,\ 42, \ldots$ (the value 6 is too small, being below $T_4 = 10$).
lengths $2, 6$: $N \equiv 1 \pmod 2$ and $N \equiv 3 \pmod 6$ give $N \equiv 3 \pmod 6$, so $21,\ 27,\ 33, \ldots$
lengths $6, 10$: $N \equiv 3 \pmod 6$ and $N \equiv 5 \pmod{10}$ give $N \equiv 15 \pmod{30}$, so $75,\ 105, \ldots$
Incompatible pairs: $2, 4$ and $4, 6$ as above, and $4, 8$, where $N \equiv 2 \pmod 4$ meets $N \equiv 4 \pmod 8$.

Pause. Whenever the pair is compatible the solutions form a single residue class modulo $\operatorname{lcm}(k, k')$, which is the Chinese remainder theorem at work. Among the incompatible pairs every one has two even lengths, and in each of them one length carries a different power of 2 from the other.

Conjecture 14 — **CRT and lcm · Stands, repairing 13**

Two lengths k, k' occur together in some N unless both are even and $v_2(k) \neq v_2(k')$, where v_2 is the exponent of the highest power of 2 dividing the number. When they are compatible, the admissible N form a single residue class modulo $\operatorname{lcm}(k, k')$ beginning at or above $\max(T_k, T_{k'})$, so the density of numbers carrying both is $1/\operatorname{lcm}(k, k')$.

TEST. The system $N \equiv T_k \pmod k$, $N \equiv T_{k'} \pmod{k'}$ is solvable exactly when the two right-hand sides agree modulo $g = \gcd(k, k')$. If both lengths are odd both residues are 0 and they agree. If k is odd and $k' = 2u$ is even then $g = \gcd(k, u)$ is odd and divides u, which is the second residue, so they agree again. If both are even, $k = 2u,\ k' = 2v$, the residues u and v must agree modulo $2\gcd(u, v)$, and writing $u = gu',\ v = gv'$ this says $u' - v'$ is even, which for coprime u', v' means both are odd, that is $v_2(u) = v_2(v)$.

A brute-force check over all pairs below 24 agrees, and the pair $(6, 10)$ gives $u' = 3,\ v' = 5$, both odd, matching $N = 75$.

NOTICE

Write $N = 2^j m$ and list all the lengths:
45 ($j = 0,\ m = 45$): odd lengths $3, 5, 9$ (and 1), even lengths $2, 6$. 75 ($j = 0$): odd $3, 5$, even $2, 6, 10$.
10 ($j = 1,\ m = 5$): even length 4. 36 ($j = 2,\ m = 9$): odd length 3, even length 8. 60 ($j = 2,\ m = 15$): odd $3, 5$, even 8.
120 ($j = 3,\ m = 15$): odd $3, 5, 15$, and no even length at all.

Pause. The even lengths $2, 6, 10$ are $2 \cdot 1,\ 2 \cdot 3,\ 2 \cdot 5$, the lengths 4, 8 and 8 in the middle line are $4 \cdot 1$ and $8 \cdot 1$, and the odd lengths all divide the primitive part, so the question is what links the power of 2 in the even length to the number itself.

Conjecture 15 — **2-adic valuation · Stands**

In $N = 2^j m$ every odd-length chain has length dividing m, and every even-length chain has length $2^{j+1}e$ with e an odd divisor of m. Each odd divisor $d > 1$ of m produces exactly one of these, of length $\min\bigl(d,\ 2^{j+1}m/d\bigr)$.

TEST. In the factor pair $2N = k \cdot m'$ of Conjecture 5 the whole factor 2^{j+1} sits on one side, so the even factor is 2^{j+1} times an odd divisor, and the odd factor is itself an odd divisor. The smaller factor is the length, which gives the minimum above. This is why even lengths in the same N can never differ in their power of 2, and so it explains the clause in Conjecture 14. The mechanism that Conjecture 4 found by cancellation is the same factor pair seen from the other side.

NOTICE

Longest chains: $28 = 1 + \ldots + 7$ has length 7, 21 has 6, 15 has 5, 10 has 4, 6 has 3.
These are triangular numbers, and the largest k with $T_k \leq N$ is exactly the length. For the numbers in between, $N = 7$ has $T_3 = 6 \leq 7$ and $N = 11$ has $T_4 = 10 \leq 11$.

Pause. In the numbers chosen, the longest chain began at 1 and its length was the largest k for which $T_k \le N$, and the question is whether that has to be so for every N.

Conjecture 16 — Specialise · Falls

The longest chain of N has length equal to the largest k with $T_k \le N$.

TEST. For $N = 7$ the largest such k is 3, but $7 \equiv 1 \pmod 3$ fails Conjecture 12 and the only chain is $3 + 4$, of length 2. For $N = 11$ the value is 4, but $11 \equiv 3 \pmod 4$ is not 2, and the chain is $5 + 6$. Numbers $12, 13, 17, 18, 19, 22, \ldots$ also fail, and a power of 2 has no chain whatever.

Conjecture 17 — Congruence class · Stands, repairing 16

Let $L(N)$ be the length of the longest chain and k^* the largest k with $T_k \le N$. Then $L(N) \le k^* < \sqrt{2N}$, with $L(N) = k^*$ exactly when $N \equiv T_{k^*} \pmod{k^*}$, and a chain starts at 1 exactly when N is triangular.

TEST. A chain of length k needs $T_k \le N$, which bounds every length by k^*, and equality is Conjecture 12 applied to k^*. Since $k < m'$ and $km' = 2N$ in Conjecture 5 we also have $k < \sqrt{2N}$. A start at $a = 1$ is the case $N = T_k$, so $45 = T_9$ has the chain $1 + \cdots + 9$, and $\sqrt{90} \approx 9.49$ sits just above its length.

NOTICE

Shortest chains: $6 \to 3$, $12 \to 3$ $(3{+}4{+}5)$, $24 \to 3$, $30 \to 3$, $40 \to 5$ $(6{+}\ldots{+}10)$, $20 \to 5$, $28 \to 7$ $(1{+}\ldots{+}7)$. Each is the smallest odd prime factor of N. But 9 has $4 + 5$, of length 2, 5 has $2 + 3$, 10 has $1{+}2{+}3{+}4$ of length 4, and 14 has $2{+}3{+}4{+}5$ of length 4.

Pause. The first line invites a guess about the smallest odd prime factor, and the second line lists the numbers that disturb it, where it helps to compare the size of the prime with the size of the power of 2 in N.

Conjecture 18 — Specialise · Falls

The shortest chain of N has length equal to the smallest odd prime factor of N.

TEST. For $N = 9$ the smallest odd prime factor is 3, yet $4 + 5$ has length 2, and every odd N has a chain of length 2 by Conjecture 1. For $N = 10$ the prime is 5, but $1 + 2 + 3 + 4$ has length 4, below it.

Conjecture 19 — 2-adic valuation · Stands, repairing 18

If $m > 1$ and p is the smallest odd prime factor of m, the shortest chain of $N = 2^j m$ has length $\min\big(p,\ 2^{j+1}\big)$.

TEST. By Conjecture 15 each odd divisor $d > 1$ gives a length $\min(d, 2^{j+1}m/d)$, which is at least $\min(p, 2^{j+1})$ because $d \ge p$ and $2^{j+1}m/d \ge 2^{j+1}$. The bound is attained by $d = p$ when $p < 2^{j+1}$, and by $d = m$ when $p > 2^{j+1}$, the case $p = 2^{j+1}$ being impossible by parity. Thus 14 gives $\min(7, 4) = 4$, 20 gives $\min(5, 8) = 5$ and 28 gives $\min(7, 8) = 7$, and an odd N with $j = 0$ gives $\min(p, 2) = 2$.

NOTICE

45: lengths $2, 3, 5, 6, 9$ (odd $3, 5, 9$; even $2, 6$) 75: lengths $2, 3, 5, 6, 10$ (odd $3, 5$; even $2, 6, 10$)
3: length 2 only $5, 7, 10, 11, 13, 14$: a single even-length chain each.

Pause. In the first column odd lengths are in the majority, in 75 the even lengths win, and in the primes there are no odd lengths at all, so it matters whether the one-term chain $N = N$ is allowed to count.

Conjecture 20 — Specialise · Falls

Counting only genuine chains of two or more terms, the chains of even length never outnumber those of odd length.

TEST. Every odd prime is a counterexample, since $3 = 1 + 2$ has one even-length chain and none of odd length, and 75 has three even and two odd.

Conjecture 21 — Pair off · Stands, repairing 20

Counting the one-term chain N as a chain of odd length 1, the number of even-length chains is at most the number of odd-length chains. A chain from the odd divisor d has odd length precisely when $d < \sqrt{2N}$.

Test. An even-length chain comes from an odd divisor $d > \sqrt{2N}$ and the pairing $d \mapsto N/d$ sends it to an odd divisor below $\sqrt{N/2} < \sqrt{2N}$, which yields an odd-length chain, and distinct d give distinct images. For 75 the divisors $d = 15, 25, 75$ map to $5, 3, 1$, so the three even chains of lengths $10, 6, 2$ are matched by the odd chains of lengths $5, 3, 1$, with the trivial chain providing the third partner.

IV · EXTENDING THE QUESTION

Integers of any sign, units, and the average

NOTICE

3: allowing zero and negatives, $[3]$, $[1,2]$, $[0,1,2]$, $[-2,\dots,3]$. (4 runs; $\tau_{\text{odd}}(3) = 2$)
9: $[9]$, $[4,5]$, $[2,3,4]$, $[-1,\dots,4]$, $[-3,\dots,5]$, $[-8,\dots,9]$. (6 runs; $\tau_{\text{odd}}(9) = 3$)

Pause. In each case the runs that are not entirely positive are mirror images of the positive ones under cancellation, and the count is twice $\tau_{\text{odd}}(N)$.

Conjecture 22 — Pair off · Stands

Allowing consecutive integers of any sign and length at least 1, N has exactly $2\,\tau_{\text{odd}}(N)$ such runs.

Test. The pair (k, m') of Conjecture 5 and the reversed pair (m', k) both give runs once the requirement $k < m'$ is dropped, and the reversed pair gives a run of length m' starting at $a = (k - m' + 1)/2 \le 0$. The pairing of Conjecture 4 is exactly this reflection. A brute-force count agrees for every $N < 80$.

NOTICE

Counting chains with a coprimality condition: $N = kF$ has a centred chain of length k, and we ask for $\gcd(k, F) = 1$. For $k = 3$ these are $N = 6, 12, 15, 21, 24, \dots$ (so $F = 2, 4, 5, 7, 8$), for $k = 5$ they are $N = 15,\ 20,\ 30,\ 35,\ 40,\ 45, \dots$, while $9 = 3 \cdot 3$ is excluded from the first list because $\gcd(3, 3) = 3$.
Computed: up to $X = 10^5$ the number of N with such a chain of length k is $22221,\ 15998,\ 7405,\ 3551,\ 2714$ for $k = 3, 5, 9, 15, 21$.

Pause. The condition $\gcd(k, F) = 1$ asks for F to be a unit modulo k, and the number of residues mod k that are units is Euler's $\varphi(k)$, so that is the first place a totient is natural. It says that k divides N *exactly*, with no further factors of the primes of k left over in N/k.

Conjecture 23 — Totient · Stands

The number of $N \le X$ for which an odd $k \ge 3$ is the length of a centred chain with $\gcd(k, N/k) = 1$ is asymptotically $X\varphi(k)/k^2$. Splitting $F = N/k$ by $g = \gcd(F, k)$ recovers the total X/k through Gauss's identity $\sum_{d \mid k} \varphi(d) = k$.

Test. Among any k consecutive values of F exactly $\varphi(k)$ are coprime to k, and F runs up to X/k, giving $\frac{X}{k} \cdot \frac{\varphi(k)}{k}$. The computed counts above are $22221,\ 15998,\ 7405,\ 3551,\ 2714$ against $22222,\ 16000,\ 7407,\ 3556,\ 2721$ from the formula. Chains themselves count divisors of N and never needed units, which explains why the totient stayed absent from the first twenty-two conjectures; it enters as soon as a question asks *how* a divisor divides, and the sum $\sum \varphi(d) = k$ is the same identity that splits all multiples of k by their gcd with k.

NOTICE

Average number of chains for $N \le X$: 4.0292 at $X = 10^4$, 5.1803 at 10^5, 6.3316 at 10^6, rising by $1.1513 = \frac{1}{2}\ln 10$ per decade.
Counting pairs (a, k) instead of numbers: each length $k \ge 2$ contributes $\lfloor (X - T_{k-1})/k \rfloor$ starts a, and lengths run up to about $\sqrt{2X}$.

Pause. There are two ways to count chains, one by divisors of each N and one by lattice points (a, k) under a boundary, and the second route lets the harmonic sum $\sum 1/k$ produce a logarithm together with a constant that can be written down.

Conjecture 24 — **Lattice count · Stands numerically**

The mean degeneracy over $N \le X$ is $\frac{1}{2}\ln X + \frac{1}{2}\ln 2 + \gamma - \frac{3}{2} + O(X^{-1/2})$, which is $\frac{1}{2}\ln X - 0.5762$, where γ is Euler's constant.

TEST. The total number of chains with $N \le X$ is the number of pairs (a, k), $a \ge 1,\ k \ge 2$, with $ka + T_{k-1} \le X$, which is about $\sum_{k=2}^{K}\big(X - T_{k-1}\big)/k$ with $K \approx \sqrt{2X}$.

The sum of X/k is $X(\ln K + \gamma - 1)$ and the sum of $T_{k-1}/k = (k-1)/2$ is $K^2/4 = X/2$, so the total is $X\big(\frac{1}{2}\ln X + \frac{1}{2}\ln 2 + \gamma - \frac{3}{2}\big)$, and rounding each floor costs $O(\sqrt{X})$. The prediction $4.0290,\ 5.1803,\ 6.3315$ matches the computed $4.0292,\ 5.1803,\ 6.3316$.

THE FORM OF A NUMBER AND ITS CHAINS

Write $N = 2^j m$ with m primitive. Then N has chains if and only if $m > 1$. The chains correspond to the odd divisors $d > 1$ of m, so there are $\tau_{\mathrm{odd}}(m) - 1$ of them, and the chain for d has length $\min(d,\ 2^{j+1}m/d)$ and is centred on N/d when that length is d. A chain of length k exists when $N \ge T_k$ and $N \equiv T_k \pmod{k}$. The shortest has length $\min(p, 2^{j+1})$ and the longest is below $\sqrt{2N}$.

Habit of mind	Where it earned its place
Specialise and test	1, 2, 8, 16, 18, 20 (and every falling conjecture)
Factor and count divisors	3, 6, 7, 9, 10
Pair off by a symmetry	4, 11, 21, 22
Double and factor	5
Congruence class	12, 13, 17
CRT and lcm	14
2-adic valuation	15, 19
Totient	23
Lattice-point count	24

A companion student learner can be found at: chains_learner_ladder

[1] John Mason, with Leone Burton and Kaye Stacey, *Thinking Mathematically*, Addison-Wesley (1982), the Consecutive Sums investigation in the chapter on conjecturing, which supplies the seed problem and the examples $9 = 2+3+4$, $11 = 5+6$, $18 = 3+4+5+6$.

ADDENDUM · CONJECTURE 24 · CHAINS OF CONSECUTIVE NUMBERS

Why Euler's γ Appears

How the harmonic sum enters a count of consecutive-sum chains, and what each constant in the average is paying for

A *chain* of N is a sum of two or more consecutive positive integers equal to N, such as $15 = 4+5+6$, and $r(N)$ is the number of chains of N. Conjecture 24 says that the average of $r(N)$ over $N \le X$ is

$$\bar{r}(X) \;=\; \tfrac{1}{2}\ln X + \tfrac{1}{2}\ln 2 + \gamma - \tfrac{3}{2} + O(X^{-1/2}),$$

where $\gamma = 0.5772\ldots$ is Euler's constant. The constant is unusual in a problem that contains no logarithms or limits in its statement, and its presence has a single source, which is that chains of each length k occupy a fixed proportion $1/k$ of the integers, so that adding up the lengths adds up the reciprocals $1/k$.

1 · THE DENSITY OF EACH LENGTH

Reciprocals arrive through congruences

Sliding a chain of length k one place to the right raises each of its k terms by 1 and therefore its sum by k, and the leftmost chain $1+2+\cdots+k$ has sum $T_k = k(k+1)/2$. The numbers carrying a chain of length k are accordingly $T_k,\ T_k+k,\ T_k+2k,\ldots$, a single residue class modulo k that begins at T_k. Among the integers up to X about X/k of them carry such a chain, since one integer in every k lies in the class. The total number of chains with sum at most X is the sum over lengths of these counts, and the lengths run from 2 up to the largest K with $T_K \le X$, which is about $K \approx \sqrt{2X}$. The sum of X/k over $k = 2,\ldots,K$ is a harmonic sum, so the question of what $\sum 1/k$ does is the whole question.

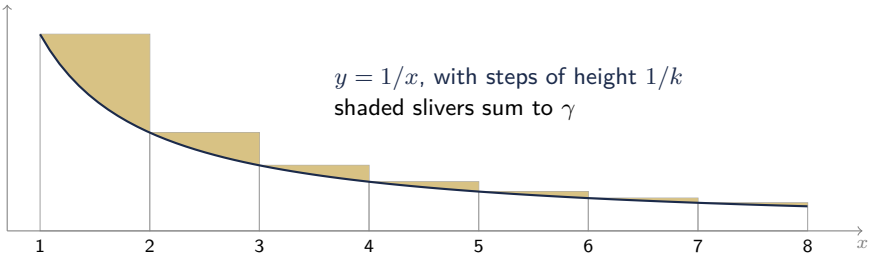
2 · WHAT THE HARMONIC SUM DOES

A logarithm plus the area left between the steps

The harmonic number $H_K = 1 + \frac{1}{2} + \cdots + \frac{1}{K}$ grows like $\ln K$, because $\ln K = \int_1^K dx/x$ is the area under the curve $1/x$. The sum is the area under a staircase of rectangles of height $1/k$ and width 1, and each rectangle overshoots the curve by a thin sliver. The slivers are shaded in the figure, and since the kth has area $\frac{1}{k} - \ln\frac{k+1}{k}$, which shrinks like $\frac{1}{2k^2}$, their total converges to a finite number. That number is γ:

$$H_K - \ln(K+1) \;=\; \sum_{k=1}^{K}\Big(\tfrac{1}{k} - \ln\tfrac{k+1}{k}\Big) \;\longrightarrow\; \gamma, \qquad \text{so} \qquad H_K = \ln K + \gamma + O(1/K).$$

The constant γ is therefore the total area of the slivers, the amount by which a sum of reciprocals exceeds the integral that approximates it, and the staircase alone fixes its value.



3 · ASSEMBLING THE CONSTANT

Four corrections, each with a source

A chain with length k and start a has sum $ka + T_{k-1}$, so the number of chains with sum at most X is

$$\sum_{k=2}^{K}\Big\lfloor \frac{X - T_{k-1}}{k} \Big\rfloor \;\approx\; X\sum_{k=2}^{K}\frac{1}{k} \;-\; \sum_{k=2}^{K}\frac{k-1}{2} \;-\; \tfrac{1}{2}K.$$

The first sum is $X(H_K - 1)$, where the -1 removes the excluded length $k = 1$. The second sum is $K(K-1)/4 \approx K^2/4 = X/2$, because $K^2 \approx 2X$. The last term collects the rounding of the floors, which loses about $\frac{1}{2}$ per length on average and is $O(\sqrt{X})$ in total. Dividing by X and using $\ln K = \frac{1}{2}\ln X + \frac{1}{2}\ln 2$ gives the stated average.

Where each term in the average comes from

$\frac{1}{2}\ln X$	The harmonic sum runs to $K \approx \sqrt{2X}$, and $\ln\sqrt{X} = \frac{1}{2}\ln X$.
$\frac{1}{2}\ln 2$	The cutoff is $\sqrt{2X}$ and the factor 2 is the 2 in $T_k = k(k+1)/2$, so $\ln\sqrt{2} = \frac{1}{2}\ln 2$.
$+\gamma$	The area of the slivers between the steps $1/k$ and the curve $1/x$ in the harmonic sum.
-1	The length $k = 1$ is not a chain, and it would have contributed $X/1$.
$-\frac{1}{2}$	The offset $T_{k-1} \approx k^2/2$ in the sum of a chain, which removes $\sum(k-1)/2 \approx X/2$.

4 · THE EVIDENCE

The data return γ when asked for it

Solving the average for the unknown constant gives $\gamma \approx \bar{r}(X) - \frac{1}{2}\ln(2X) + \frac{3}{2}$, which can be evaluated from a direct count of the chains of every $N \le X$.

X	computed $\bar{r}(X)$	predicted	difference	γ read off
10^3	2.8790	2.8777	+0.0013	0.5785
10^4	4.0292	4.0290	+0.0002	0.5775
10^5	5.1803	5.1803	0.0000	0.5772
10^6	6.3316	6.3315	+0.0000	0.5773

The difference shrinks at about the rate $X^{-1/2}$ that the rounding of the floors predicts, and leaving γ out of the prediction would miss every row by 0.577. Equivalently, the mean of $\tau_{\mathrm{odd}}(N) = r(N) + 1$ is $\frac{1}{2}\ln X + \frac{1}{2}\ln 2 + \gamma - \frac{1}{2}$, the odd-divisor counterpart of Dirichlet's classical average $\ln X + 2\gamma - 1$ for the full divisor count, which also arrives through $\sum 1/k$.

Why the lattice route and not the divisor route

Counting chains by their divisors needs the sum $\sum_{d\ \mathrm{odd} \le X} \lfloor X/d \rfloor$, which has X terms each rounded down, so the accumulated rounding is a constant fraction of X and must be tracked exactly. Counting pairs (start, length) shortens the range to $\sqrt{2X}$ terms, so the rounding costs only $O(\sqrt{X})$ and the constant falls out cleanly. The two routes agree, but only the second makes it plain that γ is the signature of the harmonic sum rather than of divisors.

5 · CHECK IT YOURSELF

A few lines of Python

```
from math import log
gamma = 0.5772156649015329
X = 10**6
t = [0]*(X+1)                 # t[n] = number of odd divisors of n
for d in range(1, X+1, 2):
    for j in range(d, X+1, d):
        t[j] += 1
mean_r = sum(t[1:])/X - 1     # r(N) = (odd divisors) - 1
print(mean_r, 0.5*log(X) + 0.5*log(2) + gamma - 1.5)
print(mean_r - 0.5*log(2*X) + 1.5)    # should be close to gamma
```

The expansion $H_K = \ln K + \gamma + O(1/K)$ is exact, while the rounding estimate $O(\sqrt{X})$ for the floors and the replacement of K by $\sqrt{2X}$ are standard arguments that have been confirmed numerically here and have not been carried through as a full proof.

ADDENDUM · THE CODE · CHAINS OF CONSECUTIVE NUMBERS

Seeing the Chains

A tour of the Colab notebook that draws five pictures of consecutive-sum chains, and how to read each one

A *chain* of N is a sum of two or more consecutive positive integers equal to N, such as $15 = 4 + 5 + 6$. Its *length* is the number of terms, and $r(N)$, the degeneracy, is the number of chains N has. The notebook `chartingConsecutiveNumbers.ipynb` is available at `colab.research.google.com/drive/1tDdNsVC_...` and turns the conjectures of the main piece into five figures, each aimed at one of them.

1 · RUNNING IT

One cell, five figures, a few seconds

The notebook is a single code cell. Choose *Runtime* → *Run all*, and it prints `wrote figures to figs` after about five seconds, with the five images saved in a folder named `figs`, which can be opened from the file browser on the left of Colab. It needs only `numpy` and `matplotlib`, both preinstalled there.

The script was written to run from a terminal, where a flag `--dark` switches the palette to a dark background and `--out` chooses the folder. Colab's kernel passes its own hidden flags to Python, which makes an ordinary `parse_args()` fail, so the notebook reads

```
args = ap.parse_args(args=[])     # ignore the kernel's flags
```

and the defaults apply (light palette, folder `figs`). To get the dark set, change that line to `ap.parse_args(args=["--dark"])`.

2 · ANATOMY

What each block of the cell does

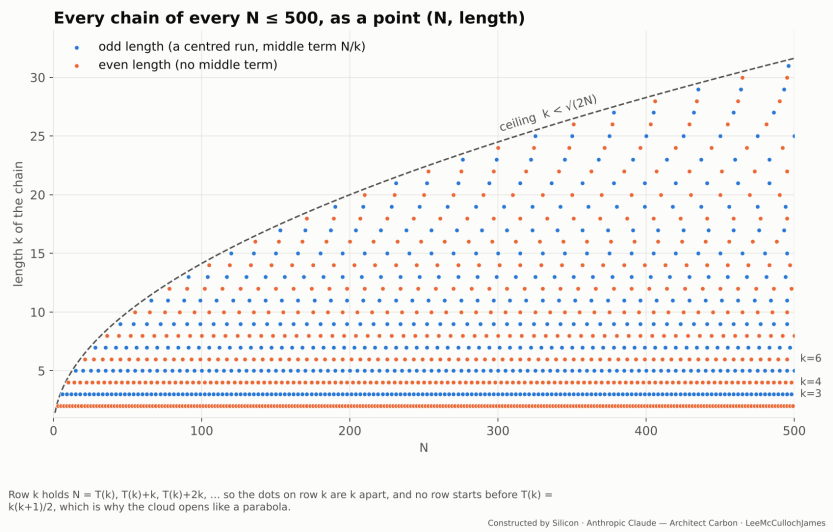
The cell from top to bottom

Palette	Two dictionaries `C`, light and dark, hold the background, ink, grid, blue, orange and gray, plus a six-step blue ramp turned into the colormap `SEQ`. Blue and orange are the two categories used throughout, and gray de-emphasises whatever is not the point of a figure.
`finish`	Every figure ends by calling this helper, which fits the layout, writes a wrapped caption and the footer stamp near the bottom edge, and saves a 170-dpi PNG.
Data	For each $N \le 5000$ a loop tries lengths $k = 2, 3, \ldots$ while $T_k = k(k+1)/2 \le N$. Taking $\mathrm{rem} = N - k(k-1)/2$, a chain of length k exists exactly when k divides rem, and its first term is rem/k. The result is the dictionary `chains`, from N to its list of (start, length) pairs.
Derived	`r` is the degeneracy (the length of each list), `lengths` the sorted lengths, `v2` the power of 2 dividing a number, and `sq` a perfect-square test.

The data loop is the only mathematics in the notebook, and it is the length law of the main piece in four lines: a chain of length k exists when $N \ge T_k$ and $N \equiv T_k \pmod{k}$. Everything after it is drawing.

3 · FIGURE 1

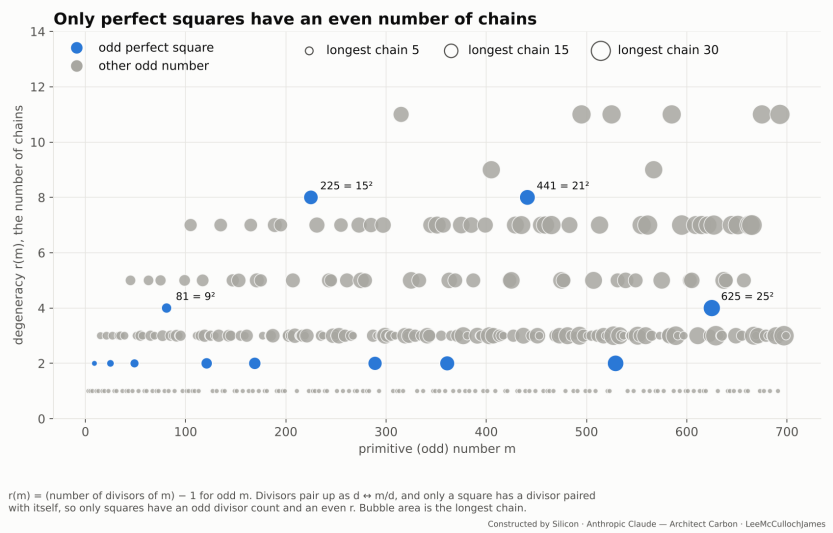
Scatter: every chain is a point



The horizontal axis is N and the vertical axis is the chain length, so each point is one chain. Odd lengths are blue and even lengths orange. Two structures stand out. The dots on each row k sit exactly k apart, because $N \equiv T_k \pmod{k}$ is a single residue class, and the cloud opens like a parabola because row k cannot begin before $N = T_k$, which is the same as the dashed ceiling $k < \sqrt{2N}$. Rows of small k are dense and rows of large k are sparse, which is the picture of density $1/k$ that Conjecture 24 sums.

4 · FIGURE 2

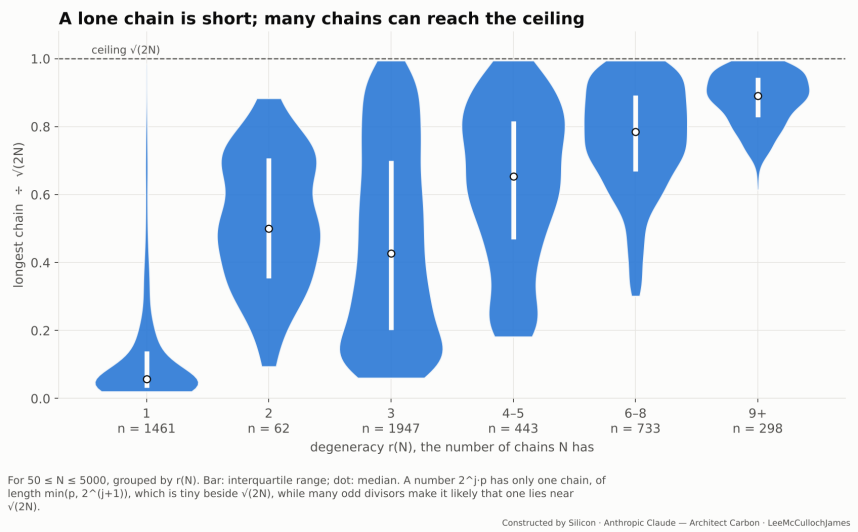
Bubbles: degeneracy of the odd numbers



Each bubble is an odd number $m \le 700$. Its height is $r(m)$, its area is the length of its longest chain, and perfect squares are drawn in blue. Since $r(m) = \tau(m) - 1$ for odd m, and the divisors of m pair up as $d \leftrightarrow m/d$ except for a square root that pairs with itself, r is even only for squares. The blue bubbles are the only ones on even heights, so the parity law of the main piece can be read from the colours. The sizes show a second fact, that high degeneracy does not guarantee a long chain.

5 · FIGURE 3

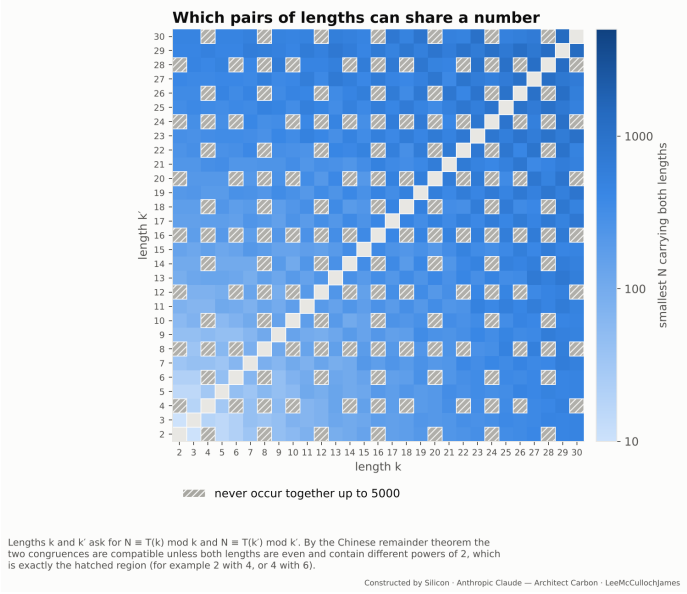
Violins: how close the longest chain gets to its ceiling



For $50 \le N \le 5000$ the longest chain is divided by $\sqrt{2N}$, so that a value near 1 means the longest chain nearly reaches the ceiling, and the numbers are grouped by $r(N)$, with the group size n under each label. A number with one chain has a short one (of length $\min(p, 2^{j+1})$), so the left violin sits near zero, while the groups with many chains spread upward because many divisors make it likely that one of them lies close to $\sqrt{2N}$. The bar is the interquartile range and the white dot the median.

6 · FIGURE 4

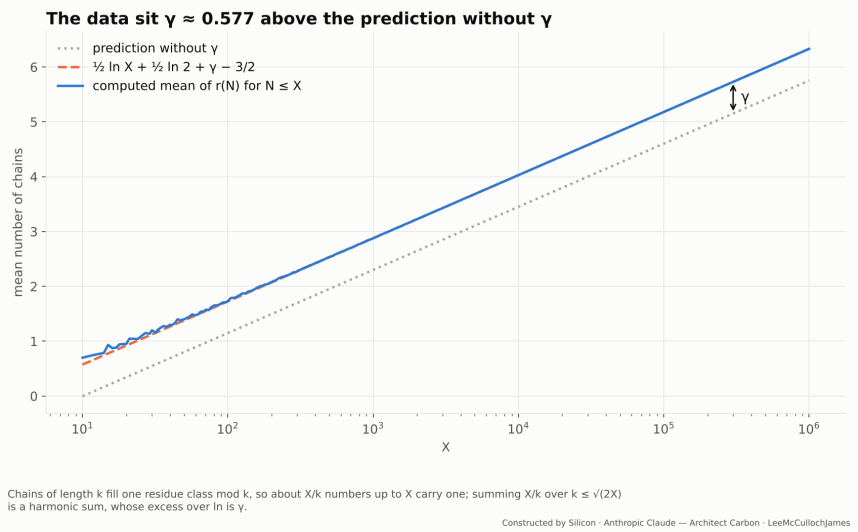
Heatmap: which lengths can share a number



Entry (k, k') is the logarithm of the smallest N that has chains of both lengths k and k'. By the Chinese remainder theorem the two congruences $N \equiv T_k \pmod{k}$ and $N \equiv T_{k'} \pmod{k'}$ can be met together unless they disagree modulo $\gcd(k, k')$, and that happens exactly when both lengths are even with different powers of 2. These pairs, such as 2 with 4 or 4 with 6, are the hatched gray squares, and no N up to 5000 carries both. Every other pair lights up, with larger lengths first meeting at larger N.

7 · FIGURE 5

Running mean: Euler's constant on a log axis



The only computation here is a sieve. An array `t` accumulates, for each odd d, a 1 at every multiple of d, so `t[n]` is the number of odd divisors of n and $r(n) = \texttt{t[n]} - 1$ up to $X = 10^6$. The cumulative sum gives the mean of r over $N \le X$, which is drawn against $\frac{1}{2}\ln X + \frac{1}{2}\ln 2 + \gamma - \frac{3}{2}$ and against the same curve with γ removed. The data hug the first curve and sit a fixed gap $\gamma \approx 0.577$ above the second, and the gap is marked with an arrow at $X = 3 \cdot 10^5$. The origin of that gap is the subject of the earlier addendum.

Things to try

Raise `Nmax` in the scatter code to 2000 and watch the rows fill in. Replace the odd range in the bubble code by `range(2,700,2)` and confirm that r is unchanged under doubling, which stacks each even number on the height of its odd part. Change the violin bins to split $r = 1$ by whether $N = 2^j$ or $N = 2^j p$. In the heatmap, extend `K` to $2, \ldots, 60$ and check that every hatched square still has two even lengths with different powers of 2. Set `XM` in the last block to 10^7 and check that the gap to the no-γ curve stays at 0.577.